

1 Brief Project Description

Given a molecule’s 3-D atomic coordinates and element identities as inputs, this project predicts $I(q) \in \mathbb{R}^{|Q|}$, its X-ray scattering intensity curve, as *output* (Fig. 1). $I(q)$ is a 1-D fingerprint of a molecule’s 3-D shape, used throughout structural biology and materials science. The exact calculation, the Debye equation, sums every pairwise interatomic interaction and costs $O(N^2)$ in atom count N ; the fastest closed-form reduction (Stuhrmann’s spherical-harmonic decomposition, $O(N \cdot L^2)$) is what generates the ground-truth labels used here, following the same spherical-harmonic approach as CRYSOLE. Both are non-differentiable analytic sums, not something that adapts to problem-specific structure. A graph neural network, in the tradition of SchNet and directional message passing for molecular property prediction, is well suited for three reasons: (1) atoms and their pairwise interactions map naturally onto a graph’s nodes and message passing, (2) message passing gives a learned $O(N)$ -per-round surrogate for the same underlying physics, cheap enough to sit inside a differentiable pipeline (e.g. structure refinement by gradient descent on predicted $I(q)$), and (3) the mapping from geometry to scattering curve is smooth and highly structured, which is exactly what a GNN’s inductive bias exploits.

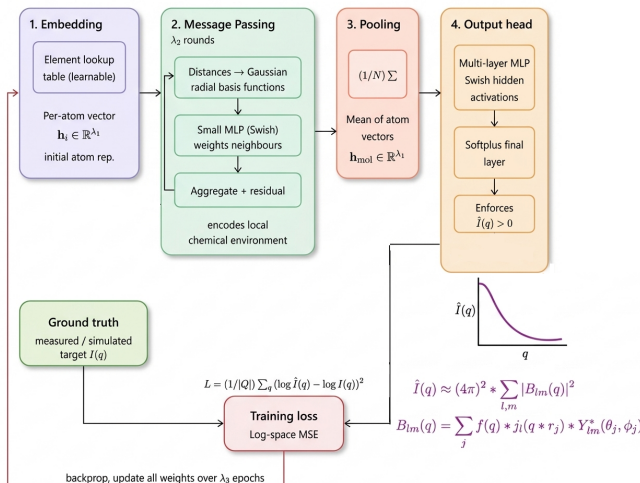


Figure 1: Input (atomic coordinates + element types) $\rightarrow$ ScatterNet $\rightarrow$ output $\hat{I}(q)$.

2 Notable Contribution

2.1 Data Processing

A dataset of 1,044,583 molecules spanning 1–6,046 atoms was assembled from 13 sources (Table 1), covering 474 unique atom/ion types (charge-resolved, e.g. 0 vs. 02-). Raw structures (.cif, .pdb, .xyz) were converted to a uniform Cartesian XYZ representation; dummy/non-physical sites were removed, deuterium was mapped to hydrogen, zero-atom structures were discarded, and non-element placeholders (X, M, Cn) were excluded from the ionic makeup.

$I(q)$ ground truth was computed from the cleaned XYZ via the Stuhrmann decomposition at $L=50$ and stored, together with the coordinates, in a compressed HDF5 database (ZFP lossless for floating-point arrays, Bitshuffle+Zstd for metadata).

To avoid padding waste across a 1–6,046 atom size range, molecules are grouped by size into 57 atom-count buckets (Fig. 2); each bucket is packed into batches capped at a fixed atom budget, splitting further if it overflows that cap, and each bucket is stratified 70/15/15 into train/val/test on its own, so every split contains both small and large molecules.

Table 1: Cleaned dataset composition by source.

Group	Molecules	Atom range
COD (Crystallography Open Database)	532,302	1–6,032
QM9 (small organic molecules)	133,844	3–29
tmQM (transition metal complexes)	108,541	7–569
rcsb_sml (PDB, small)	96,158	28–6,036
viro3D (viral protein structures)	60,488	173–6,046
hydration_shells (water solvation)	48,571	3–147
rcsb_med (PDB, medium)	31,749	2,996–6,046
mofs (metal-organic frameworks)	30,863	10–5,760
monoatomic clusters (6 metals)	1,282	2–380
binary alloy clusters	371	2–1,482
Si/Ge clusters	217	4–60
Ar/Ne clusters	127	2–55
(NaCl) _n Cl [−] clusters	70	3–71
Total	1,044,583	1–6,046

57 batcher buckets (green) vs. 5 baseline diagnostic bins (dotted); 1–10 atoms: 0 of 1,015 baseline test molecules

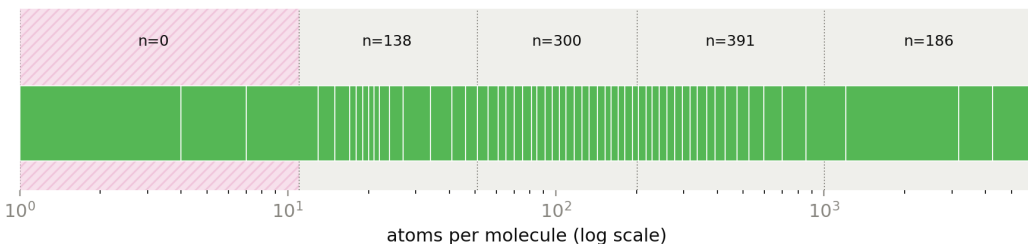


Figure 2: The 57 training buckets (green) against the 5 coarse atom-count bins used to report baseline error by molecule size. The pink hatched bin ($n=0$) marks the 1–10 atom range, where the baseline test slice happens to draw zero molecules, even though the training buckets themselves extend down to single atoms; the other four bins (shaded grey) each contain the baseline test molecules counted below their label.

There were many challenges with constructing the dataset due to data size, numerous power outages that corrupted runs, and the weeks it took to calculate all the data.

2.2 Baseline Model

Nine non-GNN baselines were implemented: closed-form physics models fit per-molecule (Guinier, Guinier-Porod, and a binned Debye sum evaluated over the histogrammed pairwise-distance distribution $P(r)$, which recovers the exact Debye equation as the bin count grows), stochastic estimators that subsample the $O(N^2)$ pairwise sum (propoEst, stratEst), and learned baselines with no access to 3-D geometry (a training-mean-by-atom-count lookup, a linear SVM over element-composition features, and a 2-hidden-layer MLP over composition + size). Fig. 3 reports the seven lowest-MSLE baselines (Guinier and Nearest-Neighbour diverge and are omitted for readability).

Binned Debye is the strongest baseline (MSLE 0.0053) since it is the closest coarse-graining of the exact ground-truth formula, and is a useful feasibility ceiling rather than a competitor to beat; the more informative comparison is against baselines that, like the primary model, only see coordinates implicitly or not at all. **Beating this baseline proves the model uses genuine 3-D structure, not just size or composition:** the composition-only baselines – Atom-Count, Linear SVM, and MLP – are the lower bound, none of them accessing coordinates at all. **Challenges:** stratEst segfaults on monoatomic molecules and both stochastic estimators are noisy enough at small sample budgets to occasionally beat their own asymptotic error, so their numbers should be read as a range rather than a point estimate.

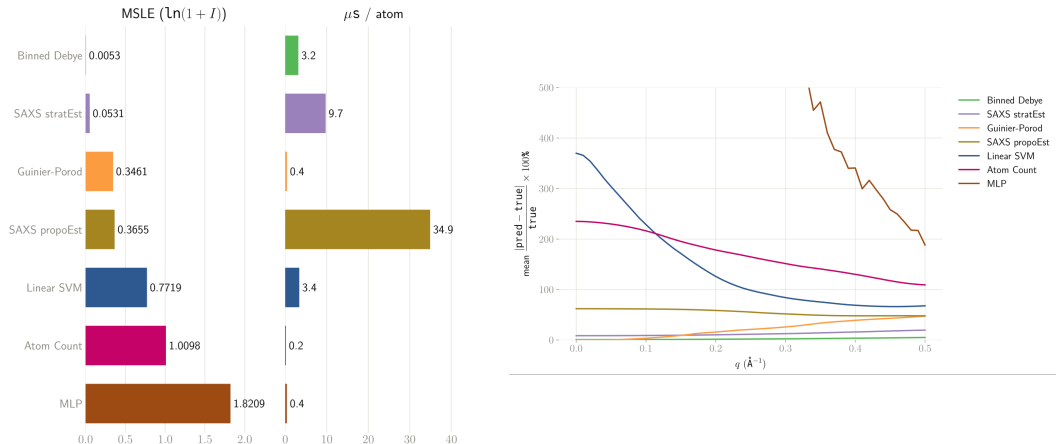


Figure 3: Quantitative (left): MSLE and $\mu S / \text{atom}$, sorted by MSLE. Qualitative (right): per- q mean percent error, baselines with explicit pairwise-distance information (Binned Debye, stratEst, propoEst) hold up best across the q -range; composition-only baselines (Atom-Count, Linear SVM, MLP) degrade earliest, since fine structural shape rather than overall size or composition determines $I(q)$ at mid-to-high q .

2.3 Primary Model

ScatterNet (Fig. 1) has four stages. **Embed** looks up a learned $\lambda_1=128$ -d vector per atom, decomposed into a complex form factor magnitude $|f_m(q)| = \text{hypot}(f_0 + f_1, f_2)$ (Cromer-Mann convention) and a bilinear kernel bandwidth $\sigma_m(q)$. **MessagePass** ($\lambda_2=3$ rounds) approximates the Gaussian kernel $k(\tilde{r}_i, \tilde{r}_j) = \exp(-\|\tilde{r}_i - \tilde{r}_j\|^2/2)$, coordinates scaled by $\tilde{r}_m(q) = r_m/\sigma_m(q)$, via $\lambda_5=64$ Random Fourier Features, reducing all-pairs aggregation from $O(M^2)$ to $O(M\lambda_5)$ per q -point. **OutputHead** is a $\lambda_4=4$ -step halving MLP at width $\lambda_3=128$ with a Softplus output, enforcing $\hat{I}(q) > 0$. The full model has **427,352 parameters**, trained with a Kratky-weighted log-scale MSE plus a form-factor penalty ($\lambda_6=0.1$) anchoring $\hat{f}_m(q)$ to the tabulated xraydb reference and a q^2 -weighted L2 penalty on σ ($\lambda_7=0.1$) preventing kernel-bandwidth blowup, via Adam, fp16, and tensor/data-parallel routing across $2 \times \text{T4 GPUs}$ (README §7, §12).

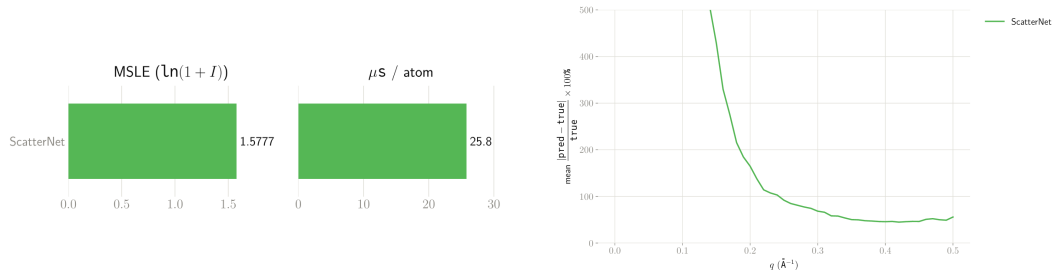


Figure 4: ScatterNet, epoch 2 of 22 (dataset_frac = 0.62): quantitative summary (left) and per- q mean percent error (right).

At epoch 2/22 (Fig. 4), ScatterNet reaches MSLE = 1.578 at 25.8 $\mu S / \text{atom}$ inference, above every composition-only baseline (Fig. 3) but still behind Linear SVM and the distance-aware physics/stochastic baselines, despite having seen only $2/22 \times 0.62 \approx 5.6\%$ of the total training signal budget. Error is concentrated in two places: percent error stays under 60% for $q > 0.2 \text{\AA}^{-1}$ but blows up at low q (the large-scale/Guinier-region signal needs the longest-range message passing to converge), and per-molecule MSLE falls monotonically with atom count (10.7 for 1–10 atoms vs. 0.35 for 1001+). **Feasibility:** loss is still falling steeply within epoch 2, and the percent-error gap sits exactly where the physics baselines predicted (Guinier region, Fig. 3) – evidence that continued training, not an architecture change, is the primary path to closing the gap with the Binned Debye ceiling. **Challenges:** tensor-parallel routing initially recompiled on every eval-mode call due to shape mismatches with train-mode buckets; fixed by keying the DP/TP routing decision on bucket shape alone (README §7).